Fairness and Transparency in Music Recommender Systems: Improvements for Artists

Karlijn Dinnissen
k.dinnissen@uu.nl
Utrecht University
Utrecht, The Netherlands

ABSTRACT

Music streaming services have become one of the main sources of music consumption in the last decade, with recommender systems playing a crucial role. Since these systems partially determine which songs listeners hear, they significantly influence the artists behind the music. However, when assessing the performance and fairness of music recommender systems, the perspectives of artists and others working in the music industry are often overlooked. Additionally, artists express a desire for greater transparency regarding why certain songs are recommended while others are not. This research project adopts a multi-stakeholder approach to close the gap between music recommender systems and the artists whose music they recommend. First, we gather insights from artists and music industry professionals through interviews and questionnaires. Building on those insights, we then aim to improve matching between end users and music from lesser-known artists by generating rich item and user representations. Results will be evaluated both quantitatively and qualitatively. Lastly, we plan to effectively communicate music recommender system fairness by increasing transparency for both end users and artists.

CCS CONCEPTS

• **Information systems** → **Recommender systems**; • **Human-centered computing**;

KEYWORDS

Music Recommender Systems, Fairness, Transparency, Human-Centered Computing

ACM Reference Format:
Karlijn Dinnissen. 2024. Fairness and Transparency in Music Recommender Systems: Improvements for Artists. In *18th ACM Conference on Recommender Systems (RecSys '24), October 14–18, 2024, Bari, Italy*. ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3640457.3688024

1 INTRODUCTION

In today's digital landscape, music recommender systems (MRSs) are ubiquitous. As these systems determine which music is recommended, they influence the decisions of end users, thereby affecting artists' chances of success [54]. Traditionally, though, research in recommender systems (RSs) has mainly focused on optimizing accuracy, defining success by how well recommended items align with inferred end user objectives. The values and goals of item providers—specifically, music artists in the context of MRSs—often remain overlooked. This work aims to ensure that MRSs better represent such values, e.g., fair representation and transparency, thereby extending beyond accuracy as a goal.

RSs research frequently focuses on specific (technical) components using offline evaluations, while remaining largely disconnected from a real-world setting. In contrast, we take a more holistic approach by considering the broader context of the music industry and its various stakeholders beyond end users. A crucial part of this approach is identifying the specific values and desired improvements of these stakeholders. Previous work in this direction identifies several key areas: providing a fair chance for each artist, enhancing insight into and control over the systems, and offering more diverse recommendations across various aspects [12, 26]. Yet, the body of work that directly engages with artists and other related stakeholders remains limited and preliminary. We therefore set out to further identify current issues, asking: **What do music providers and related stakeholders consider to be the main areas for fairness improvement in streaming services, and in music recommender systems in particular? (RQ1)**

Once we understand which improvements are desired, we may look to the broader RS community for approaches of mitigating item provider unfairness, such as re-ranking (post-processing) and enriching data (pre-processing) [23]. To date, only a limited number of these strategies have been applied in the context of the music domain [25, 41, 42, 47]. To address its unique challenges, these approaches must be adapted and evaluated within this specific domain. As a preliminary step in that direction, we perform a literature review focused on identifying how to address the main issues from RQ1: **Which (existing or novel) fairness improvement approaches are suitable for MRSs, taking into consideration RQ1's main points? (RQ2)**

Based on insights from our user studies (RQ1) and literature review (RQ2), we zoom in on one common issue in the music domain: lesser-known artists struggling to have their music recommended by MRSs. MRSs often rely on user-item interactions for training, where interaction with new and long-tail items is limited. This may lead to under-representation of such items in recommendations [2, 11]. To address this challenge, it may be possible to leverage textual item content, such as metadata and lyrics, to promote lesser-known artists. This approach has shown promise in other domains [6, 67]. Our corresponding research question is: **How can music item and consumer representations be leveraged**

RecSys '24, October 14–18, 2024, Bari, Italy

ACM ISBN 979-8-4007-0505-2/24/10
https://doi.org/10.1145/3640457.3688024

to promote lesser-known artists in a recommender system? (RQ3)

Enhancing fairness, and specifically the perception of fairness in MRSs, also presents several challenges that extend beyond algorithmic optimization. A critical issue is the lack of transparency within these systems, hindering artists from understanding why, when, and to whom their music is recommended. End users lack such transparency as well, which limits their ability to make informed decisions, and leaves them unaware of how their preferences impact artist fairness and the diversity of recommendations they receive [62]. Therefore, the second method with which we choose to address MRS fairness aims to improve the insight that end users have into their recommendations on several fairness aspects: **How does giving users more transparency into and control over their music recommendations, on fairness aspects in specific, impact fairness within the system? (RQ4)**

Finally, artists also express a strong desire for transparency and a willingness to engage with MRS insights if available [25]. While some studies have used explanations to improve transparency in MRSs for users (e.g., [5, 44]), to the best of our knowledge, no research has addressed providing such explanations for artists. When designing and implementing fairer MRSs, however, end users and artists agree this is vital [26, 62]. Consequently, we focus on how to make fairness-related MRS improvements perceivable to artists, enhancing their awareness and meeting their needs for transparency and agency. Our final research question is: **How can we make improvements aimed to enhance artist fairness in MRS, such as those from RQ3 and RQ4, transparent for artists? (RQ5)**

2 RELATED WORK

Here, we cover the main previous works we will build on. For each topic, we first describe research from the general RS community, and then zoom in on music domain-specific research. We conclude by going into each research gap as outlined in the Introduction.

2.1 Multi-stakeholder Music Recommendation

In the applied RS space, systems are generally designed to meet the goals of several stakeholders [3, 10, 57]. Firstly, *users* may rely on a RS to find, e.g., results or items that best fit their current needs. Secondly, the party offering the results or items (i.e., the *platform* or *service*) often has other distinctive needs, such as generating revenue from sales or subscriptions. RSs research often (in)directly addresses primarily these stakeholders' goals. Additionally, in many domains, there is a third group—item or content creators—who have their own goals and values. Such *item providers* may directly depend, e.g., for their income, on how well their items are performing on online platforms that employ RSs. Finally, there may be *indirect stakeholders* that are more indirectly affected by RSs.

Several works suggest that all impacted members of society should be considered throughout the RS development process: from design and goal selection, to implementation, and finally evaluation [29, 30, 45]. Reflecting this, research in the music domain is increasingly shifting towards including perspectives beyond just the user's. Two main works here focus on interviewing artists [26, 60]. In Ferraro et al. [26], fairness issues such as historical bias (e.g., based on country or gender), and limited opportunities for less popular artists are identified as main areas for improvement. Additionally, the interviewed artists view MRSs as greatly lacking transparency. Addressing these concerns, the remainder of this work focuses on both fairness and transparency.

We also expand the stakeholder view to those indirectly impacted by MRSs in the music industry, such as artist managers, promoters, and music producers. Their perspectives have not been explored in previous work, while these professionals may be as dependent on the outcomes of MRSs as music artists are.

2.2 Fairness of Recommender Systems

The concept of fairness is increasingly gaining attention in AI contexts, including RSs [22]. Its definition is highly dependent on each RS application, as are the issues that may lead to unfairness. However, several common sources of bias that can contribute to RS unfairness can be identified [13, 28]. For instance, historical bias may be inherent in the data, such as gender and locality imbalances [25, 47, 59]. Bias can also be introduced during data collection and model design [13]. When training a RS based on interaction data, additional challenges, such as popularity bias, may emerge [33, 41]. Such bias, often referred to as a 'the rich get richer' effect, occurs when already popular items are recommended more frequently due to greater prevalence in previous interactions or feedback [2, 11]. A cold start problem may also arise when there is insufficient information on new items [73]. When recommending items to users, position bias can surface [48]. Lastly, optimizing solely on user relevance may further negatively affect fairness, highlighting the importance of considering both users and item providers in the evaluation of these systems [41].

As potential sources of bias and unfairness are manifold, some suggest focusing on reducing causes of bias and unfairness in particular parts of a system rather than attempting to achieve fairness in the system as a whole [28, 39]. To develop a good understanding of fairness in a given context, recent literature (e.g. [49, 58]) emphasizes that it is crucial to take a multidisciplinary point of view and study the perspectives of the various stakeholders involved and affected. Generally, previous works distinguish between a fair system for users, and for item providers. Similar to RSs research as a whole, a large part of HCI research has only addressed fairness from the perspective of the end user [43]. Some exceptions of fairness *evaluations* from the item provider perspective, though, evaluate both user and item fairness [15, 68], focus on popularity bias [33], or discuss with item providers how to best evaluate whether a system is treating them fairly [61]. Other RS works have addressed *improving* fairness for item providers. A promising research avenue here is taking the multi-stakeholder nature of RSs into account by balancing multiple objectives, including artist fairness. Some works approach this by implementing bandit-based frameworks [7, 31, 41], while others explore Natural Language Processing-inspired transformer-based models [9], or other multi-objective algorithmic approaches [46, 51].

While these fairness improvement-focused works mainly apply in-processing interventions, such as adjusting models, or post-processing techniques, such as re-ranking, *pre-processing* has been largely unexplored [23]. One exception here is representation learning for RS users and items. As such representations can be based on item (meta)data in various forms (e.g., audio, text, images), they can often be generated even for users and items with little interaction data available. Using such representation as additional RS features may help mitigate issues such as popularity bias and the cold start problem [35, 36, 52]. We highlight one promising research direction emerging from the advancements in Large Language Models (LLMs): using these models to generate richer representations of items and user histories [53, 67]. For example, LLMs can generate comprehensive summaries or descriptive tags that capture broader trends, which may be richer than existing metadata. This has recently been demonstrated in the movie domain [6].

Studies analyzing fairness specifically from the music provider perspective are limited. Before the start of our work, only two studies directly addressed artists. Ferraro et al. [26] asked Spanish-speaking artists to voice their perspectives on fairness in MRS and how it might be improved. Siles et al. [60] inquired about the impact of music streaming services, and especially of curated playlists offered on those services, on artists' careers.

Work in the music domain aiming to mitigate item provider fairness issues is limited as well, though interesting work has been carried out countering gender imbalance [25, 42] and addressing item popularity bias [12, 33, 41]. The aforementioned approach of using an LLM to generate item and user representations for downstream use in a MRS has yet to be explored.

2.3 Transparency of Recommender Systems

RSs have many characteristics that affect their output, making it challenging to understand their decision-making processes [5, 43, 63]. Additionally, end users often lack insight into the information a MRS has collected on them [56]. To enhance such transparency, explanations can be provided [63]. These explanations may address *why* certain items are recommended, helping both users and developers better understand the system [71]. The field of Explainable AI (XAI) dedicates the majority of works to explaining black-box models to either developers of those models or to the end users receiving recommendations [65, 71]. A common challenge in this area is balancing transparency and control with the cognitive load imposed on users [63, 70]. Despite this, evaluations in the form of user studies in this field are sparse [55].

In the context of fairness, transparency may be seen as a prerequisite that helps to assess whether a RS is fair [61, 62, 70]. End users may use such information to make decisions aligning with their own values and personalities [24]. When they perceive recommendations as fair, their satisfaction may also increase [27]. The importance of transparency is further emphasized in Sonboli et al. [62], where end users point out that claiming a RS is fair is insufficient; stakeholders need to understand the fairness objectives embedded in the system. Following these works, more research is needed regarding how to communicate such objectives, as well as on users' perceptions of such communication. This becomes especially important when addressing multiple fairness dimensions, such as promoting less popular items and aiming to improve gender balance among item providers.

Research on transparency from the item provider's perspective is limited (with some exceptions, e.g., [14, 34, 38]), and there is no work focusing on this stakeholder in the music domain. As artists generally indicate not (entirely) understanding the inner workings of MRSs in general [26], and since XAI is highly context and recipient-dependent [37], research in this area would be particularly valuable. Moreover, if item providers cannot understand or perceive the impact of changes made to a RS intended to enhance their situation, they may not fully benefit from these changes. Item providers' needs and preferences here are, again, very domain-specific. Therefore, to effectively enhance transparency for music artists in MRSs, focusing on their unique needs and situations is vital.

To continue bridging the gap between music streaming services and their item providers, we aim to improve fairness and transparency for item providers in MRSs, guided by the five research questions.

3 RESEARCH OBJECTIVES & PROGRESS

Here, we describe the research plan and methodology we envision to use. From these five research questions, RQ1 and RQ2 have been completed [16–18, 21], and where possible, materials and data have been published Open Access [1, 19, 20]. RQ3 and RQ4 are currently in progress [64], and RQ5 has been planned. The Utrecht University Science-Geosciences Ethics Review Board has reviewed and approved the proposal related to RQ1. For subsequent research questions, an additional ethical review was conducted.[1]

3.1 Artists & Industry Professionals View (RQ1)

To initiate this research, we aimed to directly engage with artists and music industry professionals to understand their perspectives on what constitutes a fair MRS, and to identify the improvements they would like to see in such systems. Our approach was twofold: first, we conducted in-depth semi-structured interviews with artists [17], and second, we conducted a questionnaire-based study with music business professionals [21].

Artist perspective. We first extended the work by Ferraro et al. [26], who interviewed 9 Spanish-speaking artists. Adopting a similar interview methodology [1], we conducted 14 interviews with Dutch-speaking artists or pairs of artists. During these interviews, we explored how they perceive the impact of current music streaming services and the RSs embedded within them. We also inquired about which potential improvements to MRSs they envision. The interviews were conducted in Dutch in a semi-structured format, which involved posing the same set of open-ended guiding questions while leaving space for artists to bring up their own ideas as well. The audio of each interview was recorded, transcribed, annotated, and finally analyzed using Qualitative Content Analysis [40].

Results. When comparing our results to those from the interviews with Spanish-speaking artists in Ferraro et al. [26], we observe several similarities in artists' answers regarding item cold start problems, popularity bias, and fostering diversity. We also highlight the following considerations from our interviews, which will guide us in addressing the upcoming RQs:

[1]The Ethics and Privacy Quick Scan of the Utrecht University Research Institute of Information and Computing Sciences classified this research as low-risk with no fuller assessment required.

(1) Artists feel that streaming services have the responsibility to foster and increase the diversity of music consumption.
(2) Artists express a desire for more diverse music recommendations that expand users' musical landscape, even across music genres.
(3) Regarding which type of music should be fostered or promoted more in MRSs, artists generally agree on music from less popular artists (i.e., counter popularity bias) and new artists (i.e., counter cold start problem). Some artists suggest promoting music from minority genders. Location is an important factor, but there is no consensus on whether a more localized or globalized recommendation approach should be in place.
(4) Artists underline the need for a multi-stakeholder approach; they wish to foster a long-term relationship with their audience, and recognize that unhappy users will not continue listening to their music, or even any music on the streaming platform at all.

After collecting these perspectives, we proceed to incorporate the viewpoints of other stakeholders in the music industry.

Broader music industry perspective. In the second phase of our research, we aimed to gather perspectives from additional stakeholders who have not been previously addressed in MRSs fairness research. These stakeholders, who may be directly or indirectly affected by MRSs, include professionals involved in various capacities, such as working on an artists' team, managing a label, or operating as a music distributor. They may also use MRSs to discover music or artists, or have a role in which artist popularity on music streaming services is an important factor, such as concert bookers for venues and festivals. To ensure a comprehensive understanding of their perspectives, we designed a questionnaire addressing the same topics explored in the artist interviews. We chose a questionnaire as opposed to in-depth interviews to facilitate more responses, as well as enable statistical analysis of the results. For data collection, we reached out to visitors of a music industry conference (Eurosonic Noorderslag[2]). At this conference, we collected data from 35 visitors.

Results. Materials and data from this study were shared (see [19, 20]) following FAIR (*"Findability, Accessibility, Interoperability, and Reusability"*) guiding principles [69]. The key findings are the following:

(1) Similar to artists, music industry professionals confirm the importance of streaming services in their daily operations.
(2) Participants emphasize the need to increase diversity and inclusion among different types of music and artists, though opinions differ on the best methods to achieve this. They also believe that streaming services, given their considerable influence in the industry, have a responsibility to advance diversity and inclusion.
(3) Regarding transparency, participants express that MRSs embedded in streaming services lack transparency for both artists and users, and advocate for increased transparency. They also suggest that streaming services should provide artists with greater control over their presence on these platforms.

These findings provide a comprehensive view of MRS fairness from the perspectives of different stakeholders and outline key areas for improvement.

3.2 Fairness Improvement Approaches (RQ2)

Our subsequent step was establishing a foundation for identifying effective methods to improve MRS fairness for item providers. As outlined in Section 2, various approaches exist, but many have yet to be assessed within a music context. In a literature review, we therefore first collected the current work on MRS fairness from multiple stakeholders' perspectives [16]. This review shows that multi-stakeholder approaches to fairness are scarce in the music domain. Among the available studies, the majority focus on defining and evaluating fairness within RSs, rather than on strategies for addressing specific instances of unfairness.

Since RQ1 identified several areas for improvement, our subsequent RQs address the points most often mentioned by participants. The first one concerns increasing fairness for lesser-known artists (either new or long-tail). Therefore, in RQ3, we investigate a pre-processing approach aimed at improving the fair representation of these artists.

3.3 Improving Lesser-Known Artist Recommendation (RQ3)

As mentioned in Section 2, popularity bias and cold start issues may cause lesser-known artists (i.e., artists whose items are new or in the long tail) to be unfairly treated. RQ2 revealed that representation learning approaches in the pre-processing stage addressed this issue with promising results in other domains. Therefore, in RQ3, we aim to enhance the visibility of lesser-known artists in MRSs through this method. While representations may be learned from various types of input data, such as text, images, or audio [72], there are two reasons for starting with text. First, with the recent advance of LLMs, there has been great progress in computational text processing and generating capabilities. Second, textual (meta)data is most readily available for large quantities of music items. Therefore, in RQ3, we apply the representation learning method introduced in Agrawal et al. [6] in the music domain and use these LLM-generated item and user profiles in a recommendation task as described in Ren et al. [53] and Wang et al. [67].

We will use the textual data that is available for our items (i.e., songs), such as lyrics, audio descriptors, and artist biographies, using text-rich data sources such as `LFM2b Lyrics Descriptor Analyses` [50]. Combining such textual data should make each representation as informative as possible. We may use additional data to enrich the set(s), similar to, e.g., Ferraro et al. [25], who enriched `LFM-1b` with artist gender information. We will use an open source LLM[3] to increase reproducibility. We prompt this LLM to retrieve a representation for i) each item in the dataset and ii) the listening history of each user, consisting of the concatenated textual (meta) data of each item present in said history. By using these representations as input features in a RS candidate generation

[2]https://esns.nl/

[3]A version of Llama; https://llama.meta.com/

task, we aim to increase recommendations of items that have little available interaction data [35].

Preliminary results. We have completed developing a domain-agnostic service that generates item and user representations. Initially, we applied this service in the book domain, where a dataset with rich textual metadata was readily available [66]. Given the promising initial results, our next step involves integrating these representations as features within a RS. We will then compare how different RS architectures leverage the generated representations to assess the approach's generalizability. To evaluate the impact of incorporating these features, particularly for recommending less popular (long-tail) items, we will employ metrics such as categorical accuracy, % of long-tail items in the top-K results, and Mean Reciprocal Rank (MRR) @K. In this context, popularity will be defined based on how often each item occurs in the interaction data, as introduced in [4]. If the results using the initial book dataset prove satisfactory, we will extend this approach to the music domain, potentially incorporating non-text-based feature representations to improve results further.

3.4 Perceived Fairness for Users (RQ4)

In this RQ, we consider the role of end users in improving fairness for artists. This idea was brought up by artists in RQ1, where they emphasized the lack of transparency of MRSs towards users—in general and into fairness dimensions specifically. Artists expect that when users are given more insight into their recommendations on such dimensions (or even: more control), they will be more equipped to make choices that align with their own values [18]. Therefore, we set out to measure user perception of artist fairness in MRSs [64], as well as their responsiveness to explanations that touch on artist fairness.

Popularity bias mitigation perception. Our initial step targeted popularity bias, a key issue identified in RQ1 and further explored in RQ3. Building on previous user perception studies [24, 27, 32], we evaluated user satisfaction and perception of personalized music recommendations generated by algorithms explicitly designed to mitigate popularity bias. We implemented item- and user-centered bias mitigation techniques to enhance fairness for artists and users, and subsequently measured how users perceived the recommendations.

Results. Our findings indicate that both item-centered and user-centered bias mitigation techniques effectively maintained user satisfaction with the recommendation lists while promoting underrepresented items. Both approaches affected user perception by reducing familiarity with the items, which resulted in the participants reporting an enhanced sense of discovery and broadening of their musical taste. This underscores the potential benefits of recommending less popular items to enhance the overall user experience.

Impact of transparency on user decisions. The second phase will involve a user study designed to investigate the influence of explanations on end users' decision-making processes. Participants will be tasked with selecting between different playlists that range from homogeneous to heterogeneous on various dimensions, such as artist genre, nationality, and gender. After making an initial selection, we provide a description of the respective dimension, after which participants may revise their choice if desired. Once the tasks are completed, we will conduct a semi-structured interview to explore their motivations and general views on MRS fairness. This study aims to provide deeper insights into the role of transparency in shaping end users' decisions within MRSs. Additionally, it will reveal how end users perceive fairness-related improvements proposed by item providers in RQ1 and implemented in RQ3 and RQ4. We will also assess which fairness aspects (e.g., popularity, nationality, gender) users deem most important. If successful, this preliminary study may be expanded with further investigations into various types of explanations, multiple MRS contexts, and functionalities that increase user agency by providing additional control over recommendations.

3.5 Transparency for Artists (RQ5)

Based on the findings from RQ1 and previous research [26], a significant challenge for music artists is the need for more transparency into MRSs. Therefore, our final research question focuses on this issue. In particular, we aim to increase insight into specific fairness information artists desire. By addressing this need, our goal is to not only improve fairness within MRSs, but also make these improvements more perceivable to the artists themselves.

We envision answering this RQ through a three-step participatory design methodology. First, we will perform a pilot study with a smaller sample of music artists. In these semi-structured interviews, we will examine the current sources of information these artists use to gain insight into MRSs, collect their additional information needs, and ask them to envision how this could practically be integrated into a music streaming service. We will pay particular attention to artists' perception of how MRSs work, and devise strategies to align these perceptions more closely with the actual functioning of the systems. During these interviews, we also aim to show wireframes of explanations and otherwise transparency-enhancing descriptions in several places in a generic music streaming platform and discuss their potential with the artists. As there is no previous work on artist-focused MRS transparency, these wireframes will be inspired by earlier research on insight into MRSs for end users [5, 44, 56], communicating fairness of RSs [70], and research from other domains that aims to improve transparency for item providers or domain experts [8, 14, 34].

Secondly, using the insights gained from this pilot study, we will design prototypes to bring artists' ideas into practice. Particular attention will be given to which explanation techniques most suit the fairness goals addressed in previous RQs. We also consider differences among artists in information needs and acceptable cognitive load [63, 70].

To complete RQ5, we will directly evaluate those prototypes in a user study with a larger sample of music artists from varying backgrounds and degrees of popularity. If possible, we will allow participants to interact with the prototype(s) using a Think Aloud protocol. Through semi-structured interviews, we will then evaluate which type of explanation or functionality best fits which type of artist and information need. Herein, we will focus on, e.g., i) how well the fairness message is being brought across, ii) how satisfied artists are with the explanations, and iii) whether more transparency into MRSs increases their sense of empowerment.

4 CONCLUSION

The objectives of various stakeholders in MRSs are receiving increased attention in research. Nonetheless, the perceptions of stakeholder groups beyond end users, and how these groups are impacted by such systems, are often only minimally addressed or even overlooked completely. Therefore, in the work outlined in this paper, we actively involve such stakeholders at multiple stages. First, we aim to understand the values and desires of several provider-side MRS stakeholders. Subsequently, we focus on improving fairness in MRSs, specifically for artists. We then explore end user perception, and end user agency to choose music that aligns with their values. Finally, we aim to give artists more insight into MRSs and the fairness objectives incorporated in such systems. Ultimately, this research may serve as a basis for increasing the satisfaction and agency of *artists* who release their music on streaming platforms while keeping *end users* satisfied and engaged, thereby encouraging such platforms to adopt these approaches.

ACKNOWLEDGMENTS

Special thanks to Hanna Hauptmann, Christine Bauer, Azin Ghazimatin and Judith Masthoff for their guidance and feedback.

REFERENCES

[1] 2023. *Amplifying Artists' Voices: Item Provider Perspectives on Influence and Fairness of Music Streaming Platforms (Supplementary Materials)*. Zenodo. https://doi.org/10.5281/zenodo.7883507

[2] Himan Abdollahpouri. 2020. *Popularity bias in recommendation: A multi-stakeholder perspective*. Ph. D. Dissertation. University of Colorado at Boulder.

[3] Himan Abdollahpouri, Gediminas Adomavicius, Robin Burke, Ido Guy, Dietmar Jannach, Toshihiro Kamishima, Jan Krasnodebski, and Luiz Pizzato. 2020. Multistakeholder recommendation: Survey and research directions. *User Modeling and User-Adapted Interaction* 30, 1 (2020), 127–158. https://doi.org/10.1007/s11257-019-09256-1

[4] Himan Abdollahpouri, Masoud Mansoury, Robin Burke, and Bamshad Mobasher. 2019. The Unfairness of Popularity Bias in Recommendation. In *Proceedings of the Workshop on Recommendation in Multi-stakeholder Environments, co-located with the 13th ACM Conference on Recommender Systems (RecSys 2019)* (Copenhagen, Denmark) *(RMSE '19)*. CEUR Workshop Proceedings, CEUR-WS.org, Article 4, 7 pages. http://ceur-ws.org/Vol-2440/paper4.pdf.

[5] Darius Afchar, Alessandro Melchiorre, Markus Schedl, Romain Hennequin, Elena Epure, and Manuel Moussallam. 2022. Explainability in Music Recommender Systems. *AI Magazine* 43, 2 (Jun. 2022), 190–208. https://doi.org/10.1002/aaai.12056

[6] Saurabh Agrawal, John Trenkle, and Jaya Kawale. 2023. Beyond Labels: Leveraging Deep Learning and LLMs for Content Metadata. In *Proceedings of the 17th ACM Conference on Recommender Systems* (Singapore, Singapore) *(RecSys '23)*. ACM, New York, NY, USA, 1. https://doi.org/10.1145/3604915.3608883

[7] Maryam Aziz, Jesse Anderton, Kevin Jamieson, Alice Wang, Hugues Bouchard, and Javed Aslam. 2022. Identifying New Podcasts with High General Appeal Using a Pure Exploration Infinitely-Armed Bandit Strategy. In *Proceedings of the 16th ACM Conference on Recommender Systems* (Seattle, WA, USA) *(RecSys '22)*. ACM, New York, NY, USA, 134–144. https://doi.org/10.1145/3523227.3546766

[8] Aditya Bhattacharya, Jeroen Ooge, Gregor Stiglic, and Katrien Verbert. 2023. Directive Explanations for Monitoring the Risk of Diabetes Onset: Introducing Directive Data-Centric Explanations and Combinations to Support What-If Explorations. In *Proceedings of the 28th International Conference on Intelligent User Interfaces* (Sydney, NSW, Australia) *(IUI '23)*. Association for Computing Machinery, New York, NY, USA, 204–219. https://doi.org/10.1145/3581641.3584075

[9] Emanuele Bugliarello, Rishabh Mehrotra, James Kirk, and Mounia Lalmas. 2022. Mostra: A Flexible Balancing Framework to Trade-off User, Artist and Platform Objectives for Music Sequencing. In *Proceedings of the ACM Web Conference 2022* (Virtual Event, Lyon, France) *(WWW '22)*. ACM, New York, NY, USA, 2936–2945. https://doi.org/10.1145/3485447.3512014

[10] Robin Burke. 2017. Multisided Fairness for Recommendation. In *Proceedings of the Workshop on Fairness, Accountability and Transparency in Machine Learning, held at KDD 2017* (Halifax, Nova Scotia, Canada) *(FAT/ML '17)*. ACM, New York, NY, USA, 1–5. https://doi.org/10.48550/arXiv.1707.00093

[11] Òscar Celma. 2010. *Music Recommendation*. Springer Berlin Heidelberg, Berlin, Heidelberg, Chapter 3, 43–85. https://doi.org/10.1007/978-3-642-13287-2_3

[12] Òscar Celma and Pedro Cano. 2008. From Hits to Niches? Or How Popular Artists Can Bias Music Recommendation and Discovery. In *Proceedings of the 2nd KDD Workshop on Large-Scale Recommender Systems and the Netflix Prize Competition* (Las Vegas, NV, USA) *(NETFLIX '08)*. ACM, New York, NY, USA, Article 5, 8 pages. https://doi.org/10.1145/1722149.1722154

[13] Jiawei Chen, Hande Dong, Xiang Wang, Fuli Feng, Meng Wang, and Xiangnan He. 2023. Bias and Debias in Recommender System: A Survey and Future Directions. *ACM Trans. Inf. Syst.* 41, 3, Article 67 (feb 2023), 39 pages. https://doi.org/10.1145/3564284

[14] Giandomenico Cornacchia, Francesco M. Donini, Fedelucio Narducci, Claudio Pomo, and Azzurra Ragone. 2021. Explanation in Multi-Stakeholder Recommendation for Enterprise Decision Support Systems. In *Advanced Information Systems Engineering Workshops*, Artem Polyvyanyy and Stefanie Rinderle-Ma (Eds.). Springer International Publishing, Cham, Germany, 39–47. https://doi.org/10.1007/978-3-030-79022-6_4

[15] Yashar Deldjoo, Vito Walter Anelli, Hamed Zamani, Alejandro Bellogín, and Tommaso Di Noia. 2021. A Flexible Framework for Evaluating User and Item Fairness in Recommender Systems. *User Modeling and User-Adapted Interaction* 31 (07 2021), 457–511. https://doi.org/10.1007/s11257-020-09285-1

[16] Karlijn Dinnissen and Christine Bauer. 2022. Fairness in Music Recommender Systems: a Stakeholder-centered Mini Review. *Frontiers in Big Data* 5, Article 913608 (2022), 9 pages. https://doi.org/10.3389/fdata.2022.913608

[17] Karlijn Dinnissen and Christine Bauer. 2023. Amplifying Artists' Voices: Item Provider Perspectives on Influence and Fairness of Music Streaming Platforms. In *Proceedings of the 31st Conference on User Modeling, Adaptation and Personalization* (Limassol, Cyprus) *(UMAP '23)*. ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3565472.3592960

[18] Karlijn Dinnissen and Christine Bauer. 2023. How Control and Transparency for Users Could Improve Artist Fairness in Music Recommender Systems. In *Proceedings of the 24th International Society for Music Information Retrieval Conference (ISMIR '23)*. ISMIR, Milan, Italy, 482–491. https://doi.org/10.5281/zenodo.10265331

[19] Karlijn Dinnissen and Christine Bauer. 2023. Questionnaire: Music Industry Professionals' View on Music Streaming Services and Recommender Systems. https://doi.org/10.5281/zenodo.8121152

[20] Karlijn Dinnissen, Isabella Saccardi, Marloes Vredenborg, and Christine Bauer. 2023. Dataset: Music Industry Professionals' Perspectives on Music Streaming Services and Recommendation. https://doi.org/10.5281/zenodo.8185736

[21] Karlijn Dinnissen, Isabella Saccardi, Marloes Vredenborg, and Christine Bauer. 2023. Looking at the FAccTs: Exploring Music Industry Professionals' Perspectives on Music Streaming Services and Recommendations. In *Proceedings of the 2nd International Conference of the ACM Greek SIGCHI Chapter* (Athens, Greece) *(CHIGREECE '23)*. ACM, New York, NY, USA, Article 25, 5 pages. https://doi.org/10.1145/3609987.3610011

[22] Michael D. Ekstrand, Robin Burke, and Fernando Diaz. 2019. Fairness and Discrimination in Retrieval and Recommendation. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (Paris, France) *(SIGIR '19)*. ACM, New York, NY, USA, 1403–1404. https://doi.org/10.1145/3331184.3331380

[23] Michael D Ekstrand, Anubrata Das, Robin Burke, and Fernando Diaz. 2022. Fairness in Information Access Systems. *Foundations and Trends® in Information Retrieval* 16, 1–2 (2022), 177 pages. https://doi.org/10.1561/1500000079

[24] Somayeh Fatahi, Mina Mousavifar, and Julita Vassileva. 2023. Investigating the effectiveness of persuasive justification messages in fair music recommender systems for users with different personality traits. In *Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization* (Limassol, Cyprus) *(UMAP '23)*. ACM, New York, NY, USA, 66–77. https://doi.org/10.1145/3565472.3592958

[25] Andrés Ferraro, Xavier Serra, and Christine Bauer. 2021. Break the Loop: Gender Imbalance in Music Recommenders. In *Proceedings of the 2021 Conference on Human Information Interaction and Retrieval* (Virtual Event) *(CHIIR '21)*. ACM, New York, NY, USA, 249–254. https://doi.org/10.1145/3406522.3446033

[26] Andrés Ferraro, Xavier Serra, and Christine Bauer. 2021. What Is Fair? Exploring the Artists' Perspective on the Fairness of Music Streaming Platforms. In *Human-Computer Interaction – INTERACT 2021: 18th IFIP TC 13 International Conference* (Bari, Italy) *(INTERACT '21, Vol. 12933)*, C. Ardito, R. Lanzilotti, A. Malizia, H. Petrie, A. Piccinno, G. Desolda, and K. Inkpen (Eds.). Springer, Cham, Germany, 562–584. https://doi.org/10.1007/978-3-030-85616-8_33

[27] Bruce Ferwerda, Michaela Berndl, Eveline Ingesson, and Markus Schedl. 2023. I Don't Care How Popular You Are! Investigating Popularity Bias in Music Recommendations from a User's Perspective. In *ACM SIGIR Conference on Human Information Interaction and Retrieval* (Austin, TX, USA) *(CHIIR '23)*. ACM, New

York, NY, USA, 357–361. https://doi.org/10.1145/3576840.3578287
[28] Ben Hutchinson and Margaret Mitchell. 2019. 50 Years of Test (Un)Fairness: Lessons for Machine Learning. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (Atlanta, GA, USA) *(FAT* '19)*. ACM, New York, NY, USA, 49–58. https://doi.org/10.1145/3287560.3287600
[29] Dietmar Jannach and Christine Bauer. 2020. Escaping the McNamara Fallacy: Towards more Impactful Recommender Systems Research. *AI Magazine* 41, 4 (Dec 2020), 79–95. https://doi.org/10.1609/aimag.v41i4.5312
[30] Dietmar Jannach, Paul Resnick, Alexander Tuzhilin, and Markus Zanker. 2016. Recommender Systems—Beyond Matrix Completion. *Commun. ACM* 59, 11 (oct 2016), 94–102. https://doi.org/10.1145/2891406
[31] Olivier Jeunen and Bart Goethals. 2021. Top-K Contextual Bandits with Equity of Exposure. In *Proceedings of the 15th ACM Conference on Recommender Systems* (Amsterdam, Netherlands) *(RecSys '21)*. ACM, New York, NY, USA, 310–320. https://doi.org/10.1145/3460231.3474248
[32] Bart P. Knijnenburg, Martijn C. Willemsen, Zeno Gantner, Hakan Soncu, and Chris Newell. 2012. Explaining the User Experience of Recommender Systems. *User Modeling and User-Adapted Interaction* 22 (2012), 441–504. https://doi.org/10.1007/s11257-011-9118-4
[33] Dominik Kowald, Markus Schedl, and Elisabeth Lex. 2020. The Unfairness of Popularity Bias in Music Recommendation: A Reproducibility Study. In *Advances in Information Retrieval* (Virtual event, Lisbon, Portugal) *(ECIR '20)*, Joemon M. Jose, Emine Yilmaz, João Magalhães, Pablo Castells, Nicola Ferro, Mário J. Silva, and Flávio Martins (Eds.). Springer International Publishing, Cham, Germany, 35–42. https://doi.org/10.1007/978-3-030-45442-5_5
[34] Markus Langer and Cornelius J. König. 2021. Introducing a multi-stakeholder perspective on opacity, transparency and strategies to reduce opacity in algorithm-based human resource management. *Human Resource Management Review* 33, Article 100881 (2021), 13 pages. https://doi.org/10.1016/j.hrmr.2021.100881
[35] Jingjing Li, Ke Lu, Zi Huang, and Heng Tao Shen. 2017. Two Birds One Stone: On both Cold-Start and Long-Tail Recommendation. In *Proceedings of the 25th ACM International Conference on Multimedia* (Mountain View, California, USA) *(MM '17)*. ACM, New York, NY, USA, 898–906. https://doi.org/10.1145/3123266.3123316
[36] Jiacheng Li, Ming Wang, Jin Li, Jinmiao Fu, Xin Shen, Jingbo Shang, and Julian McAuley. 2023. Text Is All You Need: Learning Language Representations for Sequential Recommendation. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (Long Beach, CA, USA) *(KDD '23)*. ACM, New York, NY, USA, 1258–1267. https://doi.org/10.1145/3580305.3599519
[37] Q. Vera Liao, Yunfeng Zhang, Ronny Luss, Finale Doshi-Velez, and Amit Dhurandhar. 2022. Connecting Algorithmic Research and Usage Contexts: A Perspective of Contextualized Evaluation for Explainable AI. *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing* 10, 1 (Oct. 2022), 147–159. https://doi.org/10.1609/hcomp.v10i1.21995
[38] Renkai Ma and Yubo Kou. 2023. "Defaulting to boilerplate answers, they didn't engage in a genuine conversation"': Dimensions of Transparency Design in Creator Moderation. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW1, Article 44 (apr 2023), 26 pages. https://doi.org/10.1145/3579477
[39] Masoud Mansoury, Himan Abdollahpouri, Mykola Pechenizkiy, Bamshad Mobasher, and Robin Burke. 2020. FairMatch: A Graph-Based Approach for Improving Aggregate Diversity in Recommender Systems. In *Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization* (Virtual Event, Genoa, Italy) *(UMAP '20)*. ACM, New York, NY, USA, 154–162. https://doi.org/10.1145/3340631.3394860
[40] Philipp Mayring. 2004. *Qualitative content analysis* (2 ed.). Vol. 1. Sage, 159–176.
[41] Rishabh Mehrotra, James McInerney, Hugues Bouchard, Mounia Lalmas, and Fernando Diaz. 2018. Towards a Fair Marketplace: Counterfactual Evaluation of the Trade-off between Relevance, Fairness & Satisfaction in Recommendation Systems. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management* (Torino, Italy) *(CIKM '18)*. ACM, New York, NY, USA, 2243–2251. https://doi.org/10.1145/3269206.3272027
[42] Rishabh Mehrotra, Niannan Xue, and Mounia Lalmas. 2020. Bandit Based Optimization of Multiple Objectives on a Music Streaming Platform. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (Virtual Event, CA, USA) *(KDD '20)*. ACM, New York, NY, USA, 3224–3233. https://doi.org/10.1145/3394486.3403374
[43] Silvia Milano, Mariarosaria Taddeo, and Luciano Floridi. 2020. Recommender systems and their ethical challenges. *Ai & Society* 35, 4 (2020), 957–967. https://doi.org/10.1007/s00146-020-00950-y
[44] Martijn Millecamp, Nyi Nyi Htun, Cristina Conati, and Katrien Verbert. 2019. To Explain or not to Explain: The Effects of Personal Characteristics when Explaining Music Recommendations. In *Proceedings of the 24th International Conference on Intelligent User Interfaces* (Marina del Ray, California) *(IUI '19)*. ACM, New York, NY, USA, 397–407. https://doi.org/10.1145/3301275.3302313
[45] Shira Mitchell, Eric Potash, Solon Barocas, Alexander D'Amour, and Kristian Lum. 2021. Algorithmic Fairness: Choices, Assumptions, and Definitions. *Annual Review of Statistics and Its Application* 8, 1 (2021), 141–163. https://doi.org/10.1146/annurev-statistics-042720-125902
[46] Marco Morik, Ashudeep Singh, Jessica Hong, and Thorsten Joachims. 2020. Controlling Fairness and Bias in Dynamic Learning-to-Rank. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20)*. ACM, New York, NY, USA, 429–438. https://doi.org/10.1145/3397271.3401100
[47] Ricardo S Oliveira, Caio Nóbrega, Leandro Balby Marinho, and Nazareno Andrade. 2017. A Multiobjective Music Recommendation Approach for Aspect-Based Diversification. In *Proceedings of the 18th International Society for Music Information Retrieval Conference* (Suzhou, China) *(ISMIR '17)*. ISMIR, 414–420. https://doi.org/10.5281/zenodo.1416999
[48] Zohreh Ovaisi, Ragib Ahsan, Yifan Zhang, Kathryn Vasilaky, and Elena Zheleva. 2020. Correcting for Selection Bias in Learning-to-Rank Systems. In *Proceedings of The Web Conference 2020* (Virtual Event, Taipei, Taiwan) *(WWW '20)*. ACM, New York, NY, USA, 1863–1873. https://doi.org/10.1145/3366423.3380255
[49] Tiago Palma Pagano, Rafael Bessa Loureiro, Maira Matos Araujo, Fernanda Vitoria Nascimento Lisboa, Rodrigo Matos Peixoto, Guilherme Aragao de Sousa Guimaraes, Lucas Lisboa dos Santos, Gustavo Oliveira Ramos Cruz, Ewerton Lopes Silva de Oliveira, Marco Cruz, Ingrid Winkler, and Erick Giovani Sperandio Nascimento. 2022. Bias and unfairness in machine learning models: a systematic literature review. (2022). https://doi.org/10.48550/arXiv.2202.08176 arXiv:2202.08176 [cs.LG]
[50] Emilia Parada-Cabaleiro, Maximilian Mayerl, Stefan Brandl, Marcin Skowron, Markus Schedl, Elisabeth Lex, and Eva Zangerle. 2024. *LFM2b Lyrics Descriptor Analyses.* https://doi.org/10.5281/zenodo.10938615
[51] Gourab K Patro, Arpita Biswas, Niloy Ganguly, Krishna P. Gummadi, and Abhijnan Chakraborty. 2020. FairRec: Two-Sided Fairness for Personalized Recommendations in Two-Sided Platforms. In *Proceedings of The Web Conference 2020* (Virtual Event, Taipei, Taiwan) *(WWW '20)*. ACM, New York, NY, USA, 1194–1204. https://doi.org/10.1145/3366423.3380196
[52] Zhaopeng Qiu, Xian Wu, Jingyue Gao, and Wei Fan. 2021. U-BERT: Pre-training User Representations for Improved Recommendation. *Proceedings of the AAAI Conference on Artificial Intelligence* 35, 5 (May 2021), 4320–4327. https://doi.org/10.1609/aaai.v35i5.16557
[53] Xubin Ren, Wei Wei, Lianghao Xia, Lixin Su, Suqi Cheng, Junfeng Wang, Dawei Yin, and Chao Huang. 2024. Representation Learning with Large Language Models for Recommendation. In *Proceedings of the ACM on Web Conference 2024* (Singapore, Singapore) *(WWW '24)*. ACM, New York, NY, USA, 3464–3475. https://doi.org/10.1145/3589334.3645458
[54] Francesco Ricci, Lior Rokach, and Bracha Shapira. 2022. Recommender Systems: Techniques, Applications, and Challenges. In *Recommender Systems Handbook* (3rd ed.). Springer US, New York, NY, 1–35. https://doi.org/10.1007/978-1-0716-2197-4_1
[55] Yao Rong, Tobias Leemann, Thai-Trang Nguyen, Lisa Fiedler, Peizhu Qian, Vaibhav Unhelkar, Tina Seidel, Gjergji Kasneci, and Enkelejda Kasneci. 2024. Towards Human-Centered Explainable AI: A Survey of User Studies for Model Explanations. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 4 (2024), 2104–2122. https://doi.org/10.1109/TPAMI.2023.3331846
[56] Richard Savcinsky and Ladislav Peska. 2023. SpotifyExplained: User-centric Mobile Application for Music Exploration. In *Adjunct Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization* (Limassol, Cyprus) *(UMAP '23 Adjunct)*. ACM, New York, NY, USA, 92–95. https://doi.org/10.1145/3563359.3597392
[57] Markus Schedl, Peter Knees, Brian McFee, and Dmitry Bogdanov. 2022. Music Recommendation Systems: Techniques, Use Cases, and Challenges. In *Recommender Systems Handbook* (3rd ed.), Francesco Ricci, Lior Rokach, and Bracha Shapira (Eds.). Springer US, New York, NY, USA, 927–971. https://doi.org/10.1007/978-1-0716-2197-4_24
[58] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and Abstraction in Sociotechnical Systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (Atlanta, GA, USA) *(FAT* '19)*. ACM, New York, NY, USA, 59–68. https://doi.org/10.1145/3287560.3287598
[59] Dougal Shakespeare, Lorenzo Porcaro, Emilia Gómez, and Carlos Castillo. 2020. Exploring Artist Gender Bias in Music Recommendation. In *Proceedings of the Workshops on Recommendation in Complex Scenarios and the Impact of Recommender Systems co-located with 14th ACM Conference on Recommender Systems (RecSys 2020)* (Virtual Event, 2020) *(ComplexRec-ImpactRS 2020, Vol. 2697)*. CEUR Workshop Proceedings, CEUR-WS.org, Article 1, 9 pages.
[60] Ignacio Siles, Amy Ross Arguedas, Mónica Sancho, and Ricardo Solís-Quesada. 2022. Playing Spotify's game: artists' approaches to playlisting in Latin America. *Journal of Cultural Economy* 15, 5 (2022), 17 pages. https://doi.org/10.1080/17530350.2022.2058061
[61] Jessie J. Smith, Aishwarya Satwani, Robin Burke, and Casey Fiesler. 2024. Recommend Me? Designing Fairness Metrics with Providers. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (Rio de Janeiro, Brazil) *(FAccT '24)*. ACM, New York, NY, USA, 2389–2399. https://doi.org/10.1145/3630106.3659044

[62] Nasim Sonboli, Jessie J. Smith, Florencia Cabral Berenfus, Robin Burke, and Casey Fiesler. 2021. Fairness and Transparency in Recommendation: The Users' Perspective. In *Proceedings of the 29th ACM Conference on User Modeling, Adaptation and Personalization* (Utrecht, The Netherlands) *(UMAP '21)*. ACM, New York, NY, USA, 274–279. https://doi.org/10.1145/3450613.3456835

[63] Nava Tintarev and Judith Masthoff. 2022. Beyond Explaining Single Item Recommendations. In *Recommender Systems Handbook* (3rd ed.), Francesco Ricci, Lior Rokach, and Bracha Shapira (Eds.). Springer US, New York, NY, 711–756. https://doi.org/10.1007/978-1-0716-2197-4_19

[64] Robin Ungruh, Karlijn Dinnissen, Anja Volk, Maria Soledad Pera, and Hanna Hauptmann. 2024. Putting Popularity Bias Mitigation to the Test: A User-Centric Evaluation in Music Recommenders. In *Proceedings of the Eighteenth ACM Conference on Recommender Systems* (Bari, Italy) *(RecSys '24)*. ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3640457.3688102

[65] Heidi Vainio-Pekka, Mamia Ori-Otse Agbese, Marianna Jantunen, Ville Vakkuri, Tommi Mikkonen, Rebekah Rousi, and Pekka Abrahamsson. 2023. The Role of Explainable AI in the Research Field of AI Ethics. *ACM Trans. Interact. Intell. Syst.* 13, 4, Article 26 (dec 2023), 39 pages. https://doi.org/10.1145/3599974

[66] Mengting Wan and Julian McAuley. 2018. Item recommendation on monotonic behavior chains. In *Proceedings of the 12th ACM Conference on Recommender Systems* (Vancouver, British Columbia, Canada) *(RecSys '18)*. ACM, New York, NY, USA, 86–94. https://doi.org/10.1145/3240323.3240369

[67] Jianling Wang, Haokai Lu, James Caverlee, Ed H. Chi, and Minmin Chen. 2024. Large Language Models as Data Augmenters for Cold-Start Item Recommendation. In *Companion Proceedings of the ACM on Web Conference 2024* (Singapore, Singapore) *(WWW '24)*. ACM, New York, NY, USA, 726–729. https://doi.org/10.1145/3589335.3651532

[68] Lequn Wang and Thorsten Joachims. 2021. User Fairness, Item Fairness, and Diversity for Rankings in Two-Sided Markets. In *Proceedings of the 2021 ACM SIGIR International Conference on Theory of Information Retrieval* (Virtual Event) *(SIGIR '21)*. ACM, New York, NY, USA, 23–41. https://doi.org/10.1145/3471158.3472260

[69] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, and Philip E et al. Bourne. 2016. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data* 3, 1 (2016), 9 pages. https://doi.org/10.1038/sdata.2016.18

[70] Mireia Yurrita, Tim Draws, Agathe Balayn, Dave Murray-Rust, Nava Tintarev, and Alessandro Bozzon. 2023. Disentangling Fairness Perceptions in Algorithmic Decision-Making: the Effects of Explanations, Human Oversight, and Contestability. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. ACM, New York, NY, USA, Article 134, 21 pages. https://doi.org/10.1145/3544548.3581161

[71] Yongfeng Zhang and Xu Chen. 2020. Explainable Recommendation: A Survey and New Perspectives. *Foundations and Trends® in Information Retrieval* 14, 1 (2020), 1–101. https://doi.org/10.1561/1500000066

[72] Hongyuan Zhu, Ye Niu, Di Fu, and Hao Wang. 2021. MusicBERT: A Self-supervised Learning of Music Representation. In *Proceedings of the 29th ACM International Conference on Multimedia* (Virtual Event, China) *(MM '21)*. ACM, New York, NY, USA, 3955–3963. https://doi.org/10.1145/3474085.3475576

[73] Ziwei Zhu, Jingu Kim, Trung Nguyen, Aish Fenton, and James Caverlee. 2021. Fairness among New Items in Cold Start Recommender Systems. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Virtual Event) *(SIGIR '21)*. ACM, New York, NY, USA, 767–776. https://doi.org/10.1145/3404835.3462948